

# NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

Exam: UPSC

Subject: Statistics

Questions: 27

Source: Official source PDF extraction

AI-generated PYQs: 0

## 1. [UNKNOWN] paper-i

0-  $P = \begin{pmatrix} 2 & 3 & 0 & 0 & 1 \\ p & 1 & \dots & p & 0 \end{pmatrix}$ ,  $0 < p < 1$ ,  $x = 1$  Let  $g(x) = 1$ ,  $x = 2, 3$ . If  $Y_n = n$ , show that  $\{Y_0, Y_1, \dots\}$  is not a Markov Chain (MC). 10 Derive Kolmogorov — Smirnov 10 test for two samples and illustrate. FSI-P-STSC 5 ===== PAGE 6 ===== SECTION B

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-I.pdf

## 2. [UNKNOWN] paper-i

Q1. SECTION A (a) The random variable  $X$  has the exponential probability density function (pdf) given by  $f(x) = X \exp(-Xx)$ ,  $x > 0$ ,  $X > 0$ . Show that, for any  $c > 0$ ,  $P(X > c) = \exp(-Ac)$ . Hence show that, for any  $x > c$ ,  $P(X > x | X > c) = \exp(-A(x - c))$ . Deduce the conditional pdf of  $X$  given that  $X > c$ , and comment briefly. 2+3+3 (b) 12-5% of the candidates in a specific examination of a certain year are known to have a score of at least 70% in Statistics Paper I, while another 18-1% have a score of at most 38%. Assuming the underlying distribution to be normal, estimate the probability that in a random sample of 5 such candidates, 2 will have a score of 60% or more. [You may use the following information : For a standard normal variate  $X$ ,  $P(X < K) = 0.637, 0.875$  and  $0.919$  for  $K = 0.35, 1.15$  and  $1.40$  respectively] 8 (c) The random variable  $Y$  has geometric distribution with parameter  $p$  ( $0 < p < 1$ ), i.e.,  $P(Y = y) = (1 - p)^y p$  for  $y = 0, 1, 2, \dots$ . Find the probability generating function of  $Y$ , and hence find the mean and variance of this distribution. 8 The random variables  $Y_1, Y_2, \dots, Y_n$  constitute a random sample from this distribution. Define  $Y_r = Y$ . Find an unbiased estimator of 1 (to be shown), and check for its consistency. 6+2 (d) Let  $X_1, X_2, \dots, X_n$  be a random sample from a distribution with probability density function  $f(x) = -x)^{13 - \theta}$ ,  $0 < x < 1$ , where  $\theta$  ( $> 0$ ) is an unknown parameter. Find the maximum likelihood estimator, 6, of  $\theta$ . 5 Suppose that the values of  $X_1, X_2, \dots, X_n$  are not known, but you do know  $Y$ , the number of  $X_i$  less than 0.5. State the distribution of  $Y$ . 3 FSI-F-STSC 2 ===== PAGE 3 ===== I? •

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STATISTICS\_PAPER\_I.pdf

## 3. [UNKNOWN] paper-i

Q1. (i) For  $n$  events  $A_1, A_2, \dots, A_n$ , show that  $n \cap A_i = (n - 1)$ .  $i=1$  (ii) Let  $\{A_n\}$  be an increasing sequence of sets (events), then show that  $(\lim_{n \rightarrow \infty} P(A_n)) = P(\lim_{n \rightarrow \infty} A_n) = P(\cup_{n=1}^{\infty} A_n)$ . 1/4'  $n=1$  Let  $K_2 = \{1, 2, 3, 4\}$  and  $p_i = P(M_i)$ ,  $i = 1, 2, 3, 4$ . Assume that  $P_3 = 4 - 2$ ,  $P_4 = \dots$ . Define the events  $E_1 = (1, 3)$ ,  $E_2 = (2, 3)$  and  $E_3 = (3, 4)$ . Check whether  $E_1, E_2$  and  $E_3$  are mutually independent. 8 Suppose that  $X_1, \dots, X_r$  form a random sample from a Uniform distribution on the interval  $[0_1, 0_2]$  where both  $0_1$  and  $0_2$  are unknown ( $- \infty < 0_1 < 0_2 < \infty$ ). Find the maximum likelihood estimators of  $0_1$  and  $0_2$ . 8 Suppose that  $X_1, \dots, X_n$  form a random sample from a Gamma distribution for which the value of parameter  $a$  is unknown ( $a > 0$ ) and value of parameter  $\theta$  is known. Show that the joint probability density function of  $X_1, \dots, X_n$  has a Monotone Likelihood Ratio (MLR). 8 FSI-P-STSC 2 3+5 3 and ===== PAGE 3 ===== (e) (i) Let  $\{A_n\}$  be a sequence of events such that  $P(X_n = n_2) = \dots$ ,  $n \geq 1$ . Check whether Strong Law of Large Numbers holds. Also comment about Weak Law of Large Numbers. (ii) Let  $X_n$  prob.,  $X$  and  $Y$ ; prob.  $> Y$ , then show that  $X$  is in probability. 8

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-I.pdf

## 4. [UNKNOWN] paper-i

Q2. (a) The random variables density function  $f(x, y)$  (pdf) 1  $X$  and  $Y$  are jointly distributed with probability  $P(X < y, Y < x) = 1 - 2xy$ ,  $x > 0$ , otherwise. 3log2 Find the marginal pdf of  $X$ . 4 Find the conditional pdf  $f(y/x)$ , for  $1 < x < 2$ ,  $1 < y < 2$ , and hence evaluate  $P(Y < 1.5 | X = 1.1)$ . 4+2 (b) Suppose  $X$  and  $Y$  are independent random variables having Poisson distributions with respective means  $A$ , ( $> 0$ ) and  $M$  ( $> 0$ ). Show that  $X + Y$  also follows a Poisson distribution. 5 Find  $P(X = k | X + Y = n)$ , where  $k$  and  $n$  are integers with  $0 \leq k \leq n$ . For given  $n > 0$ , name the distribution you have obtained. 5 (c) If a certain team loses one of its matches, then it has probability 0.5 of losing the next match and probability 0.4 of drawing it. If the team draws a match, then it has probability 0.3 of losing the next match and probability 0.4 of drawing it. If the team wins a match, then it has probability 0.2 of losing the next match and probability 0.4 of drawing it. Model this as a Markov chain, and write down its transition matrix. 5 If the team loses its first game of the season, find the probability that it wins its

third game. 5 (d) Suppose  $(X, Y)$  follows bivariate normal distribution  $N_2(0, 0, 1, 1, \rho)$ . Then show that  $10 \rho = \cos \theta$ , where  $\theta = \angle POX < 90^\circ$ .

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STATISTICS\_PAPER\_I.pdf

### 5. [UNKNOWN] paper-i

2) (1- $\rho$ )<sup>2</sup> 1.2.3.... P 1p.23... p - 1). Discuss the significance of this result. 8+2 Describe Lahiri's method, with justification, of drawing a sample of size  $n$  from a population of size  $N$  with probabilities proportional to the sizes of the respective units. 2+8

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STATISTICS\_PAPER\_I.pdf

### 6. [UNKNOWN] paper-i

Q2. (a) Suppose that a diagnostic test for HIV(+) status has both sensitivity ( $P(\text{Test positive} | \text{Disease})$ ) and specificity ( $P(\text{Test negative} | \text{No disease})$ ) equal to 0.95 and the real possibility ( $P(\text{Disease})$ ) is 0.005. Find the probability that a subject is truly HIV(+) given that the diagnostic test is positive. 8 (b) Let a continuous random variable  $X$  have probability density function (pdf) given by  $f(x) = \frac{1}{2} e^{-|x|}$ , otherwise 0. Find the probability density function of  $Y = \sin X$ . 10 (c) Let  $(X, Y)$  be uniformly distributed over  $R = \{(x, y) : x^2 + y^2 \leq 1\}$ . Find the distributions of  $(X | Y = y)$  and  $(Y | X = x)$ ,  $E(Y | X = x)$  and  $E(X | Y = y)$ , and Correlation coefficient  $\rho_{XY}$ . 12 FSI-P-STSC 3 ===== PAGE 4 ===== (d) Let  $X$  follow Pareto distribution with parameters  $a$  and  $\theta$  with probability density function (pdf)  $f(x) = \frac{\theta a^\theta}{x^{\theta+1}}$ ,  $x > 0$ ,  $\theta > 0$ . Then show that  $Y = \ln(-X^{-\theta})$  follows Logistic distribution. /0

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-I.pdf

### 7. [UNKNOWN] paper-i

Q3. (a) (i) State the Central Limit Theorem (CLT) for sums of independently and identically distributed random variables. (ii) Examine if CLT holds for the sequence of random variables  $(X_k)$  with  $P(X_k = i) = \frac{1}{2^k}$ ,  $k = 1, 2, \dots$  (b) The probability that a certain tomato seed germinates is  $\theta$ . A gardener sows a set of  $n$  such seeds and finds that  $x$  of them have germinated. It can be assumed that seeds germinate independently of one another. Find the posterior distribution of  $\theta$ , assuming that its prior distribution is beta with parameters  $a$  and  $p$ . FSI-P-STSC 3 4 6 10 ===== PAGE 4 ===== 1  $N(0, \sigma^2)$ , elsewhere.  $P(X_i = K)$ ,  $K = 1, 2, \dots, N$  (i) Define a family of UMAU confidence sets with a given confidence coefficient  $1 - \alpha$  for a parameter  $\theta$ . (ii) Derive a UMAU confidence set for  $\theta$  with confidence coefficient  $1 - \alpha$  in sampling from  $N(\mu, \sigma^2)$  with  $\mu$  unknown. Show that this set is actually an interval. Let  $X_1, X_2, \dots, X_n$  be a random sample from a uniform distribution  $U(0, 1)$ . Find a Uniformly Minimum Variance Unbiased Estimator (UMVUE) (to be shown) of  $N$ . 10

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STATISTICS\_PAPER\_I.pdf

### 8. [UNKNOWN] paper-i

Q3. (a) Suppose that  $x_1, x_n, \dots$  form a random sample from a Beta distribution with parameters  $a$  and  $b$  where the value of  $a$  is known and the value of  $b$  is unknown ( $b > 0$ ). Obtain a sufficient statistic for  $b$ . 8 Let  $x_1, x_n, \dots$  be a random sample from  $P(\theta)$ . Find Uniformly Minimum Variance Unbiased Estimator (UMVUE) of  $\theta$ . 9. Compute the estimator based on the following sample observations: 12 1, 3, 0, 8, 5, 6, 9, 2, 7, 5 The lifetimes of fluorescent lamps are independent exponential random variables with parameter  $\lambda$ . Suppose that  $\lambda$  has a prior distribution Gamma with parameters 4 and 20,000. After we observe 5 lamps with lifetimes 2911, 3403, 3237, 3509 and 3118 (in hours), we want to predict the lifetime  $X_6$  of the next lamp. Obtain the predictive distribution of  $X_6$ . 10 Suppose  $X_1, X_2, \dots, X_n \sim N(a, \sigma^2)$  where  $\sigma^2$  is known. Find the Likelihood Ratio Test (LRT) for  $H_0: \mu = \mu_0$  vs  $H_1: \mu > \mu_0$ . Show that the test in (i) is a UMP test. 10 FSI-P-STSC 4 ===== PAGE 5 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-I.pdf

### 9. [UNKNOWN] paper-i

Q4. (a) (i) Define a maximum likelihood estimator for a parameter  $\theta$ , and state the large sample properties of this type of estimator under regularity conditions, to be stated clearly. 8 (ii) Suppose that the number of a particular plant species in sampling quadrats follows a Poisson distribution with mean  $\lambda$  and it is required to estimate  $\theta = \lambda^2$ . A random sample of  $n$  such quadrats yields the numbers  $X_1, X_2, \dots, X_n$ . Find the unbiased estimator of  $\theta$  (to be shown), and also find the Cramer-Rao lower bound for the variance of this unbiased estimator of  $\theta$ . 6+6 Discuss how you would estimate the unknown number of fishes of a given size (say, 1 kg) in a large pond, based on hypergeometric probability model, using

catch-recatch method. Indicate how this technique can be used for estimating the unknown number of tigers in a large forest, using pug-mark method. [Hint : Pug-mark can be inflicted on a tiger from a distance when it comes for drinking in a pond. A tiger thus marked can be identified at a later time, again from a distance.] 7+3 Discuss Wald-Wolfowitz Runs test to examine if two samples of sizes  $n_1$  and  $n_2$  come from an identical population, against the alternative that the two populations from which the two samples have been taken, differ in any respect whatsoever. 10 FSI-F-STSC 4 =====  
PAGE 5 ===== SECTION B

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STATISTICS\_PAPER\_I.pdf

#### 10. [UNKNOWN] paper-i

Q4. (a) For any sequence  $\{A_n, n \geq 1\}$  of events with  $E_p(A_k) < c$ , comment on  $\lim_{k \rightarrow \infty} P(\bigcap_{n=k}^{\infty} A_n) = 0$  Suppose that an examination contains 99 questions arranged in a sequence from the easiest to the most difficult. Suppose that the probability that a particular student will answer the first question correctly is 0.99, the probability that he will answer the second question correctly is 0.98 and in general, the probability that he will answer the  $i$ th question correctly is  $1 - \frac{1}{i}$ , for  $i = 1, 2, \dots, 99$ . It is assumed that 100 all questions will be answered independently and that the student must answer at least 60 questions correctly to pass the examination What is the probability that the student will pass? (Tables 1(a) and 1(b) are provided at the end.) 10 Let  $\{X_0, X_1, \dots\}$  be a Markov Chain (MC) with transition probability matrix  $\begin{pmatrix} 1 & 2 & 3 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-I.pdf

#### 11. [UNKNOWN] paper-i

Q5. (a) Discuss the randomised response technique for estimating a sensitive parameter, such as the proportion of tax-evaders in a community. 10 Explain the terms 'Estimable Parametric Function', 'Error Function' and 'Best Linear Unbiased Estimator (BLUE)' in connection with linear estimation. Show that the sample mean in sampling from  $N(\mu, \sigma^2)$  is BLUE for  $n$ . 6+4 (c) The variance of a stratified random sample mean can be written as  $\frac{1}{n} \sum_{i=1}^k N_i^2 \frac{S_i^2}{N_i} - \frac{(\sum_{i=1}^k N_i S_i)^2}{N^2}$  Explain the (conventional) notation used here. Suppose the total cost of sampling is  $C = C_0 + C_1 n_i$ ,  $i = 1, 2, \dots, k$  where  $C_0, C_1, C_k$  are positive constants. If  $\text{Var}(\bar{Y}_{st})$  is fixed and the stratum sample sizes are chosen to minimise the total cost of sampling, show that the  $i$ th stratum sample size  $n_i$  is proportional to  $N_i S_i / C_i$ ,  $i = 1, 2, \dots, k$ . 10 (ii) State briefly three reasons why an analyst may wish to perform a principal component analysis. 5 (iii) Under what circumstances would it be sensible to use the variance-covariance matrix instead of the correlation matrix in principal component analysis? 5

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STATISTICS\_PAPER\_I.pdf

#### 12. [UNKNOWN] paper-i

Q5. (a) Let  $N$  be the incidence matrix of a Balanced Incomplete Block Design (BIBD) of order  $b \times v$ . Show that  $b \geq v$ . 8 (i) A plane is fitted to  $n = 33$  observations on  $(X_1, X_2, Y)$  and it is found that overall regression is just significant at  $\alpha = 0.05$  level. Find out  $R^2$  based on the available information. (Tables 2(a) and 2(b) are provided at the end.) (ii) In a one-way layout, show that for all values of  $i$ , and  $j$ ,  $j = 1, 2, \dots, n$ ,  $i = 1, 2, \dots, p$ ,  $w_i = w_3 = y_i$  are uncorrelated with each other (under usual assumptions). 3+5 Let  $Y$  follow  $N_2(\mu, \Sigma)$  (lit, 1.t2, cs, 4, p). Find distribution of  $Z = CY$  where  $C$  is a  $2 \times 2$  non-singular matrix. Also give an explicit form of matrix  $C$  such that  $CEC' = I$ , where  $E$  is the dispersion matrix of  $Y$ . 8 Construct a  $2^3$  design in two blocks where ABC is confounded. 8 Find the condition under which systematic sample mean is more efficient than a simple random sample mean. 8

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-I.pdf

#### 13. [UNKNOWN] paper-i

Q6. (a) Suppose  $Y_1, Y_2, Y_3, Y_4$  are independent with  $E(Y_1) = E(Y_2) = \mu_1 + \mu_2$ ,  $E(Y_3) = E(Y_4) = \mu_1 + \mu_3$ ,  $\text{Var}(Y_i) = \sigma^2$ ,  $i = 1, 2, 3, 4$ . Determine the condition of estimability of the parametric function  $\beta_1 + \beta_2 + \beta_3$ . Obtain a solution of the normal equations and the sum of squares due to error. 3+4+3 FSI-P-STSC 5 ===== PAGE 6 ===== Write in detail about the use of orthogonal polynomials in regression analysis, and how we choose its appropriate degree for a given data set. 7+3 Show that, with usual notation,

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STATISTICS\_PAPER\_I.pdf

#### 14. [UNKNOWN] paper-i

Q6. (a) Let  $X_1, X_2, \dots, X_n$  be a random sample from  $N(\mu, \sigma^2)$ . Consider the hypothesis  $H_0: \mu = \mu_0$ , where  $\mu_0$  and  $\sigma^2$  are known. Derive the MLE for  $\mu$ . Show that  $-2 \log$  likelihood ratio is  $n(\bar{X} - \mu_0)^2 / \sigma^2$ . Deduce the distribution of the statistic. (b) Suppose that a chemical engineer considers the time of reaction for a chemical process as a function of the type of catalyst used. Four catalysts are being investigated and the procedure consists of selecting a batch of raw materials. The observations recorded are as shown below: Catalyst Batch of Raw Materials

| Catalyst | Batch 1 | Batch 2 | Batch 3 | Batch 4 |
|----------|---------|---------|---------|---------|
| 1        | 73      | 75      | 68      | 72      |
| 2        | 75      | 72      | 75      | 73      |
| 3        | 73      | 74      | 71      | 75      |
| 4        | 75      | 67      | 72      | 72      |

Identify the design and analyse it. (Tables 2(a) and 2(b) are provided at the end.) Let  $X_1$  and  $X_2$  be independent random vectors of order  $(n_1 \times p)$  and  $(n_2 \times p)$  respectively and let  $n_i$  rows of  $X_i$  ( $i = 1, 2$ ) be independently and identically distributed as  $N(\mu_i, E_i)$ . Show that for  $u_i = 122$  and  $E_1 = E_2$ ,  $n_1 n_2 D_2^2 T_2(p, n - 2)$ , where  $n = n_1 + n_2$ ,  $D_2^2$  denotes sample Mahalanobis distance statistic and  $T_2$  denotes the Hotelling's  $T^2$ . Find an unbiased estimator of the population mean under probability proportional to size (PPS) sampling with replacement. Find the variance of this estimator and also give an estimator of this variance.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-I.pdf

#### 15. [UNKNOWN] paper-i

Q7. (a) Discuss the advantages of factorial designs. Give the complete analysis, including the layout, of a  $32$ -factorial design laid out in  $r$  replicates of  $3$  incomplete blocks each, totally confounding the factorial effect  $AB^2$ . Explain the concept of missing-plot technique and the analysis of an  $r \times r$  Latin Square design with one missing value, clearly stating the assumptions needed. Define a symmetrical balanced incomplete block design and show that the number of treatments common between any two blocks of a symmetrical balanced incomplete block design is a constant, say  $A$ . Show that a randomised block design is an orthogonal design.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STATISTICS\_PAPER\_I.pdf

#### 16. [UNKNOWN] paper-i

Q7. (a) In a simple linear regression problem  $Y = \mu_0 + \mu_1 X + E$ , in which a patient's response  $Y$  to a new drug  $B$  is to be related to his response  $X$  to a standard drug  $A$ . Suppose  $10$  pairs of observations  $(x_i, y_i)$ ,  $i = 1, \dots, 10$  are obtained. Determine the MLEs of  $\mu_0$ ,  $\mu_1$  and  $\sigma^2$  of their corresponding parameters,  $\mu_0$ ,  $\mu_1$  and  $\sigma^2$ . Obtain variance ( $\sigma^2$ ), variance ( $\mu_1$ ) and  $\text{corr}(\mu_0, \mu_1)$ . In a two-way layout with  $k$  observations in each cell ( $k \geq 2$ ), construct a test of the null hypothesis that all the interactions are zero. (Model and all assumptions are required to be specified in detail). For the covariance matrix given by  $\begin{pmatrix} 1 & 4 & 4 \\ 4 & 10 & 10 \\ 4 & 10 & 10 \end{pmatrix}$ , obtain the proportion of the total population variance explained by the first principal component. In simple random sampling where  $n$  paired observations  $(y_i, x_i)$ ,  $i = 1, \dots, n$  are drawn, obtain the regression estimators and derive its large sample variance.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-I.pdf

#### 17. [UNKNOWN] paper-i

Q8. (a) Define one-stage and two-stage cluster sampling. How do cluster sampling and stratified sampling differ, both in construction and use? Give an example of a survey that uses both stratification and clustering in the sample design. (b) Define Hotelling's  $T^2$  statistic, and indicate its applications. Show that  $mP F - p + 1, p + 1 P'$  symbols having their usual significance. (i) Discuss the need of ratio estimation and find the bias and the variance of the sample estimator  $t'$ , of population ratio  $R$ . (ii) Explain briefly the circumstances under which the ratio estimator of a population mean will be less precise than the sample mean of a simple random sample of the same total size. Show that an incomplete block design is connected if and only if the rank of the  $C$ -matrix is  $v - 1$ , where  $v$  denotes the number of treatments.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STATISTICS\_PAPER\_I.pdf

#### 18. [UNKNOWN] paper-ii

1. Answer all of the following: (a) Distinguish between tolerance limits and specification limits. Interpret the situation when tolerance limits are not included within the specification limits. (b) Describe the different types of redundancy and their USCS in reliability improvement. (c) Explain the following terms: (i) OC function (ii) ATI (iii) AOQL (iv) LTPD. (d) Consider the Markov chain with the following transition probability matrix:  $P = \begin{pmatrix} 0 & 1 & 2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 4 & 0 \\ 2 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$ . Is the chain irreducible? Check whether the states of the chain are periodic, persistent and non-null. (e) What is a linear programming problem (LPP)? Differentiate between the simplex method and two-phase simplex

method for solving LPP. 8 Develop the OC function of a double sampling plan for attributes. 10 Distinguish between process control and product control. What are the Statistical techniques to achieve these? 15 The following data is on minutes to breakdown for an insulating fluid. There were 11 tests at 30 kV. After 100 minutes, there were 7 breakdowns at the following times in minutes : 7-74, 17-05, 20-46, - 21-02, 22-66, 43-40, 47-30. The other 4 units had not failed. Assuming the failure time distribution is exponential calculate the WILE for the Reliability Function at time  $t = 25$  minutes. 15 3.(a) Use duality to solve the LPP : Minimize  $z = 3x_1 + x_2$  Subject to  $2x_1 + 3x_2 \geq 1$ ,  $x_1 + x_2 \geq 0$ . 20 fsi-d-stsc 2 ===== PAGE 3 ===== d's 3.(b) Solve the following transportation problem : D1 D2 D3 D4 a, 10 7 3 6 1 6 8 3 7 4 5 3 b j 2 6 4 In a game of matching coins with two players, suppose A wins one unit of value when there are two heads, wins nothing when there are two tails and loses unit of value when there is one head and one tail. Determine the pay off matrix and the optimal strategies for both the players. 20 Solve the following LPP : Maximize  $z = 2x_1 + 3x_2 + x_3$  Subject to  $-3x_1 + 2x_2 + 3x_3 = 8$ ,  $-3x_1 + 4x_2 + 2x_3 = 7$ ,  $x_j \geq 0$ , 20 SECTION 'B'

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STATISTICS\_PAPER\_II.pdf

### 19. [UNKNOWN] paper-ii

Q2. (a) Explain the meanings of (i) basic solutions, and (ii) feasible solutions in a linear programming problem with  $m$  conditions and  $n$  variables. Using simplex method, solve the following linear programming problem: 15 Maximize  $z = 3x_1 + 2x_2 + 5x_3$  subject to  $x_1 + 2x_2 + x_3 \leq 430$ ,  $3x_1 + 2x_2 \leq 460$ ,  $4x_2 \leq 420$ ,  $x_1, x_2, x_3 \geq 0$ . FSI-D-STSC 2 ===== PAGE 3 ===== (b) What are the different control charts for attributes used in Industrial Inspection of manufactured units? Also calculate the control chart for the number of defects and comment whether the process is under control or not based on the following data: 15 Piece No. : 1 2 3 4 5 6 7 8 9 10 Number of defects : 4 3 6 3 0 1 3 5 7 8 (c) Describe classification of states in a Markov chain. What is an  $n$ -step transition probability? Prove the following Chapman-Kolmogorov equation for transition probabilities :  $(n + 1) p_{jk}(n) = \sum_i p_{ji}(n) p_{ik}(1)$  where symbols have their usual meanings. 10

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-II.pdf

### 20. [UNKNOWN] paper-ii

Q3. (a) Define the terms of Reliability function and Failure rate function of a random variable denoting life time of a component. Establish the relation between them if any exist. Also prove that  $\frac{dh(t)}{dt} = -h(t)$  where  $h(t)$  is the failure rate function. 15 (b) A food company puts mango juice into cans advertised as containing 200 ml of the juice. Quantity of the cans after filling for 10 samples of 4 cans each are taken by a random method at an interval of 60 minutes. Following presented below are the excess over 200 ml in each can. Construct an  $\bar{X}$ -chart to control the volume of mango juice for filling. (Table Value of  $A_2$  for  $n = 4$  is 0.729). 15 Sample No. Can 1 Can 2 Can 3 Can 4 1 15 12 13 20 2 10 8 8 14 3 8 15 17 10 4 12 17 11 12 5 18 13 15 4 6 20 16 14 20 7 15 19 23 17 8 13 23 14 16 9 9 8 18 5 10 6 10 24 5 FSI-D-STSC 3 ===== PAGE 4 ===== (c) What do you mean by an assignment problem (AP)? Describe the steps of the method for solving an AP. A car rental service has a surplus of one car in each of the cities 1, 2, 3, 4, 5 and 6 and a deficit of one car in each of the cities 7, 8, 9, 10, 11 and

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-II.pdf

### 21. [UNKNOWN] paper-ii

Q4. (a) What is a multi-channel queueing problem? Deduce difference-differential equations for the (M/M/K) : (co/FIFO) queueing system and obtain the steady-state solution for the system size. 15 (b) Develop hazard functions when the life pattern of a system was described by 15 Exponential, Weibull, and Lognormal distribution. (c) Describe a two-person zero sum game and in this context explain the terms (i) pay-off matrix, (ii) saddle point, and (iii) mixed strategies. Let  $f(i, j)$  be a real-valued function and be defined whenever  $i \in A, j \in B$ . Suppose that both  $\max_i \min_j f(i, j)$  and  $\min_j \max_i f(i, j)$  exist. 15 Then prove that a necessary and sufficient condition that  $\max_i \min_j f(i, j) = \min_j \max_i f(i, j)$  is that the function  $f$  possesses a saddle point. 10 FSI-D-STSC 4 ===== PAGE 5 ===== SECTION B

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-II.pdf

### 22. [UNKNOWN] paper-ii

5. Answer all of the following: 8x5=40 What is homogeneity error in connection with index number? What steps will you take to reduce this error? 8 Discuss the rationale behind the moving average method in determining trend: 8 What do you mean by parallel tests? Obtain the effect of test length on the reliability of a test. 8 What are the different components of a time series? Why a multiplicative model is often preferred for explaining a time series on economic and

business data? 8 Define and compare the crude birth rate, gross and net reproduction. rates as measures of fertility. How do you determine the rate of growth of a population? 8 6.(a) Write down the usefulness of cost of living index number. Can we have a single cost of living index number for India? Give reasons in favour of your answer. 20 0, 02 03 3 5 7 3 fsi-d-stsc ===== PAGE 4 ===== The following table gives (with no missing values) the overall and groupwise cost of living indices (With the year 2000 as base) with six different expenditure groups and their respective weights, for the urban middle class people of a particular city, in 2004 and 2005 : Group Weight Group Index (2004) Group Index (2005) Food 350 117 120 Clothing and Footwear 156 113 118 Housing 187 118 — Transport and Vehicles 108 112 117 Durable household goods 76 102 111 Other items 123 121 125 Total 1000 119-5 Compute the overall cost of living index for 2004. Hence, determine the group index for housing in the year 2005. 20 In a music competition there are m participants who are to be evaluated by n judges who want to judge the participants by ordering their performances. Discuss in detail how can you combine the verdicts of the judges to come to a final conclusion. Clearly mention the assumptions, if any. 20 Given the following table for lx, the number of rabbits living at age x, complete the life table for rabbits : x 0 1 2 3 4 5 6 c 100 90 80 75 60 30 0 at least one of them will be alive for one year more. A, B, C will be alive for two years of time. Exactly one of the three is alive in two years of time. All will be dead in two years time. 20 8.(a) (i) Which scale is mainly used to measure a Psychological trait? Suppose you have two score distributions given by 0 x s. and 0 y /2 12 for scores x and y respectively. Find the equivalence curve relating x and y and mention its utility. Describe in brief, a practical method for obtaining equivalent scores from the score distributions for x and y without going into the process of graduation. 20 8.(b) Explain multicollinearity with an example. Discuss the different methods of detection of multicollinearity. 20 fsi-d-stsc 4 6.(b) If A, B, C are three rabbits of age 1, 2 and 3 years respectively, find the probability that

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STATISTICS\_PAPER\_II.pdf

### 23. [UNKNOWN] paper-ii

Q5. (a) What are the components of time series data ? Explain least square method of fitting a trend line. 8 What is meant by Identification problem in simultaneous equation models ? Distinguish between exactly identified, over identified and unidentified. 8 With usual notations, explain abridged life table columns and establish the relationship between Age Specific Death Rate ( $illx$ ) and Life Table Death Rate ( $.q_x$ ). 8 Explain the method of collecting demographic data using the method of registration, stating its uses and limitations. 8 What are T-scores and standard scores ? Mention the uses of T-scores and compare it with standard scores. 8

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-II.pdf

### 24. [UNKNOWN] paper-ii

Q6. (a) Explain time reversal and factor reversal tests. Show that the Marshall-Edgeworth index number lies in between Laspeyres' and Paasche's index numbers. 15 What is Heteroscedasticity ? Explain the following methods of detecting heteroscedasticity : (i) Graphical method, (ii) Park test, and (iii) Glejser test. 15 Explain the need of using standardized death rates. Also describe the method of computing standardized death rate using indirect method. 10

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-II.pdf

### 25. [UNKNOWN] paper-ii

Q7. (a) Explain the role of CSO and NSSO, the official statistical organisations in India, in collecting statistics. 15 Define Total Fertility Rate (TFR) and Net Reproduction Rate (NRR). Explain their importance in context of population growth. 15 Explain different phases of Box-Jenkins methodology for time series data analysis using flow chart. 10 FSI-D-STSC ===== PAGE 6 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/STAT\_PAPER-II.pdf

### 26. [UNKNOWN] paper-ii

Q8. (a) Define and distinguish between stationary and stable populations. Explain Gompertz population growth model and state your comment. 15 What is Multicollinearity ? Give indicators which help in detecting multicollinearity. Also explain any two remedial measures to minimise multicollinearity. 15 Explain the concepts of reliability and validity of scores in educational and psychological experiments. Mention the effect of lengthening a test upon reliability and validity. Test A has a reliability coefficient of 0.70 and a correlation of 0.40 with the criterion p. What would be the correlation of test A with the same criterion, if the test were tripled in length? 10 FSI-D-STSC 6

**27. [UNKNOWN] paper-ii**

12. The distance (in kilometres) between cities with a surplus and cities with a deficit are shown in the matrix below. Work out an optimal assignment of surplus cars and the corresponding total distance to be travelled.

|         | 10 | 7  | 8  | 9  | 10 | 11 |
|---------|----|----|----|----|----|----|
| 12 From | 1  | 2  | 3  | 4  | 5  | 6  |
| 41      | 72 | 39 | 52 | 25 | 51 | 22 |
| 29      | 49 | 65 | 81 | 50 | 27 | 39 |
| 60      | 51 | 32 | 32 | 45 | 50 | 48 |
| 52      | 37 | 43 | 29 | 40 | 39 | 26 |
| 30      | 33 | 82 | 40 | 60 | 51 | 30 |